

~~known.~~
with ~~known.~~

$$\int \sin x \, dx = -\cos x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\therefore \frac{d}{dx}(-\cos x + C) = \sin x$$

$$\therefore \frac{d}{dx}(\tan x + C) = \sec^2 x$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\therefore \frac{d}{dx}(\sin x + C) = \cos x$$

$$\therefore \frac{d}{dx}(-\cot x + C) = \csc^2 x$$

$$\int \sec x \cdot \tan x \, dx = \sec x + C$$

$$\int \csc x \cdot \cot x \, dx = -\csc x + C$$

$$\therefore \frac{d}{dx}(\sec x + C) = \sec x \tan x$$

$$\therefore \frac{d}{dx}(-\csc x + C) = \csc x \cdot \cot x$$

$$\int e^x \, dx = e^x + C$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C \quad \left. \begin{array}{l} a > 0 \\ a \neq 1 \end{array} \right\}$$

$$\therefore \frac{d}{dx}(e^x + C) = e^x$$

$$\therefore \frac{d}{dx}\left(\frac{a^x}{\ln a} + C\right) = a^x$$